
2nd IEEE International Conference on AI Engineering and Innovation (AIEI2027) : Submission (1532) has been created.

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Wed, Sep 30, 2026 at 11:50 PM

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Hello,

The following submission has been created.

Track Name: Track 7: AI for Education, Technology Management and Engineering

Paper ID: 1532

Paper Title: A Live Computation Index and Event-Based Routing Framework for Offline Smartphone Group AI Chat

Abstract:

If processing is to be carried out on the prompts in an AI group chat offline without the use of cloud servers, it is going to be quite difficult since a phone within the group has to process all the prompts. Phones have varying computational capacities, temperature and power status and this is not consistent during the conversation. In our research, we have designed an algorithm that incorporates the Live Computation Index (LCI) together with the event based routing algorithm to select hosts in an offline AI group chat among phones. LCI and other computational capacity models are capable of giving accurate ratings of the devices even under thermal constraints and resource congestion, whereas event based routing is a flexible selection approach that does not require reevaluation of the ratings and reaction to changes in real-time device status. Our algorithm ranks all the phones within the network according to LCI based on backlog, service time and live slowdown factor. If the current host is not able to reach the feasibility threshold or another stronger phone is available, the event based routing will move the conversation to another host phone. In order to facilitate the host selection and reassignment, Algorithm should focus on tracking just a small subset of signals (thermal, overload, disconnect, and battery). As soon as the host is assigned, all requests are transmitted from the clients via the local Wi-Fi mesh, handled by the model on the device, and returned back to the whole group. Hosts which have been assigned using our Algorithm may be applied for offline chat and edge inference services, and many more. We evaluate our Algorithm on the offline phone mesh conversation dataset with the devices operating in the environment under thermal and load pressure. With respect to the experimental data, our Algorithm provides 59.1% improvement in average response time compared to Round Robin and 12.9% compared to Best Available Phone.

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Submission Questions Response:

1. Originality
Agreement accepted
2. IEEE Paper Length and Formatting Compliance
Agreement accepted
3. Quality and Standard
Agreement accepted
4. Plagiarism and AI Policy Compliance
Agreement accepted
5. Novelty and Contribution

Prior research on mobile LLM inference is mainly about the optimization of the same model running on a single smartphone. The collaborative and edge computing techniques split the model and run inference among several devices or on edge servers. None of them considers the following offline problem for a group of people: which of their phones should be chosen to host the group chat powered by an AI chatbot? The novelty of our work is in addressing this problem as a runtime and event-driven routing problem for selecting the host of the offline chat among a group of mobile devices that communicate with each other without the cloud. We propose a Live Computation Index (LCI), a lightweight metric estimating each device's response time from the queue backlog, service time and thermal slowdown factor. It is composed of just six cheap probes (decode speed, free RAM, thermal headroom, queue availability, battery level, and network RTT) and rates phones reliably even in a case of heat and load pressure. Moreover, we propose an event-based router that is not continuously ranking phones according to any periodic schedule. It only reacts to several types of events (thermal or overloaded, disconnected state or low battery, as well as a challenger phone whose LCI is at least 8% higher). In advance of any move, an admission control is performed to ensure the new host has enough free RAM to prevent useless and dangerous handoffs. Third, we develop a complete offline system not requiring a cloud server. It uses peer-to-peer mDNS/TCP discovery over Wi-Fi mesh, performs the inference on the host phone and forwards each reply to all group members. In addition, during the host switching, chat history is copied to preserve chat integrity. Finally, we evaluate the developed offline system on five real Android phones with different performance levels from low-end to high-end, running Qwen 0.6B through llama.cpp in a hot and overloaded regime. In comparison with Round Robin strategy, it reduces the average response time by 59.1%, while the difference with the Best Available Phone strategy is 12.9%. It achieves 99.4% success rate, finishes host handoff within 142 ms with 100% handoff success rate, and keeps the host temperature up to 38.4 °C, while the static host throttles at 43.8 °C.

6. Presentation Commitment
Agreement accepted

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